

Autonomous Proxy-based LLM Query Tuning

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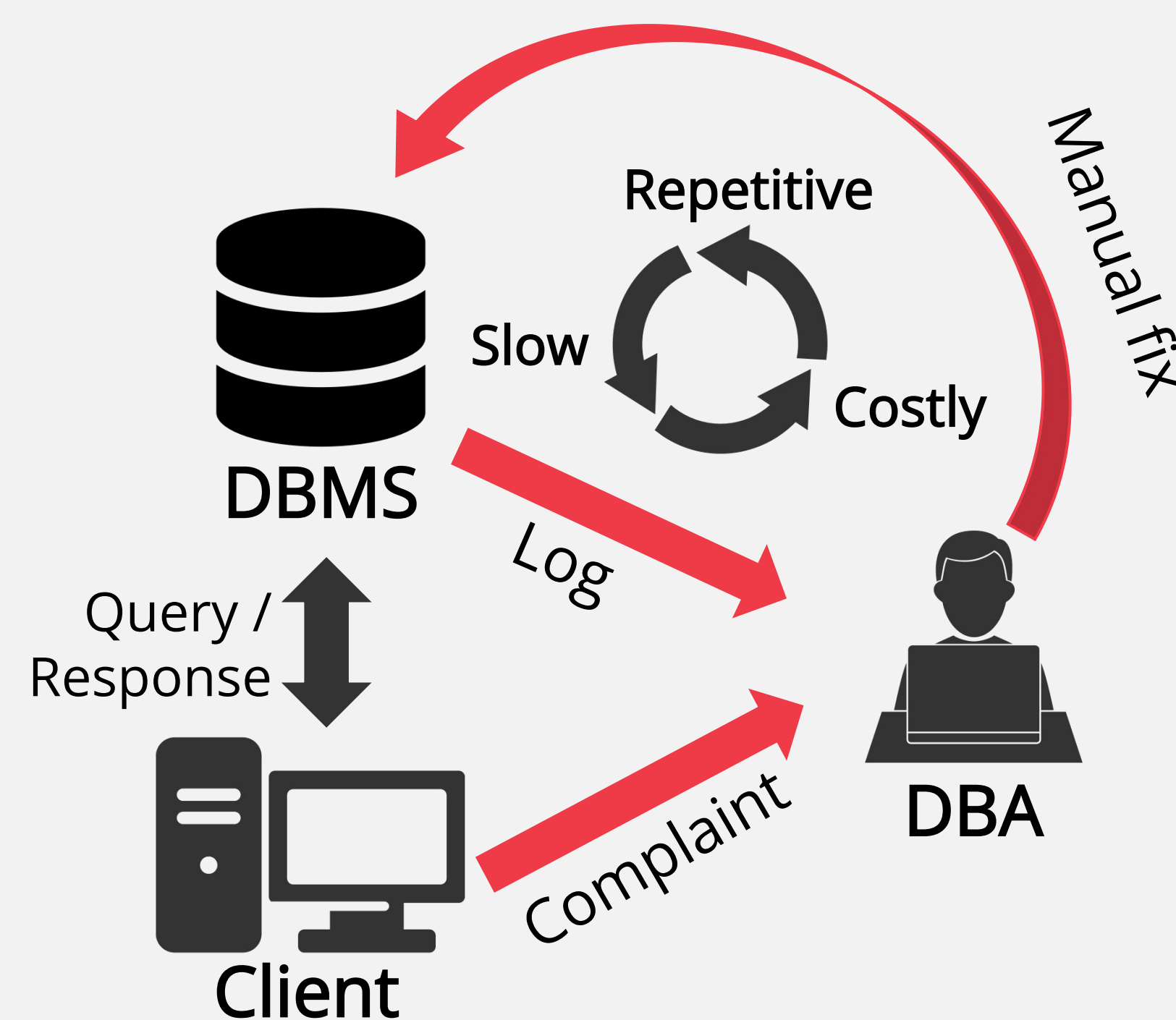


Carnegie Mellon
Database Group

Motivation

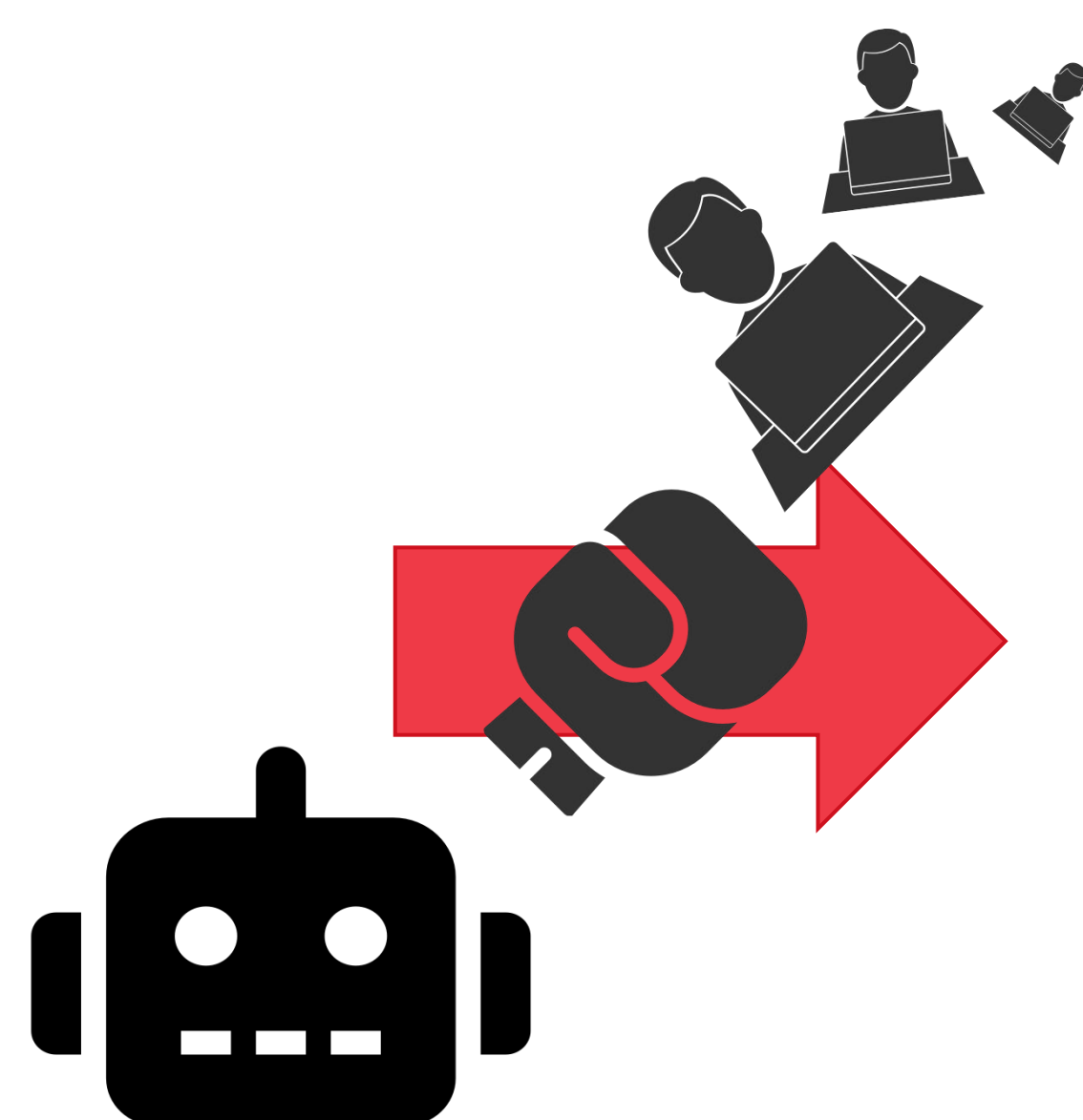
Problem

DBMS tuning relies on human expertise:
(1) What to do? (2) How to do it?



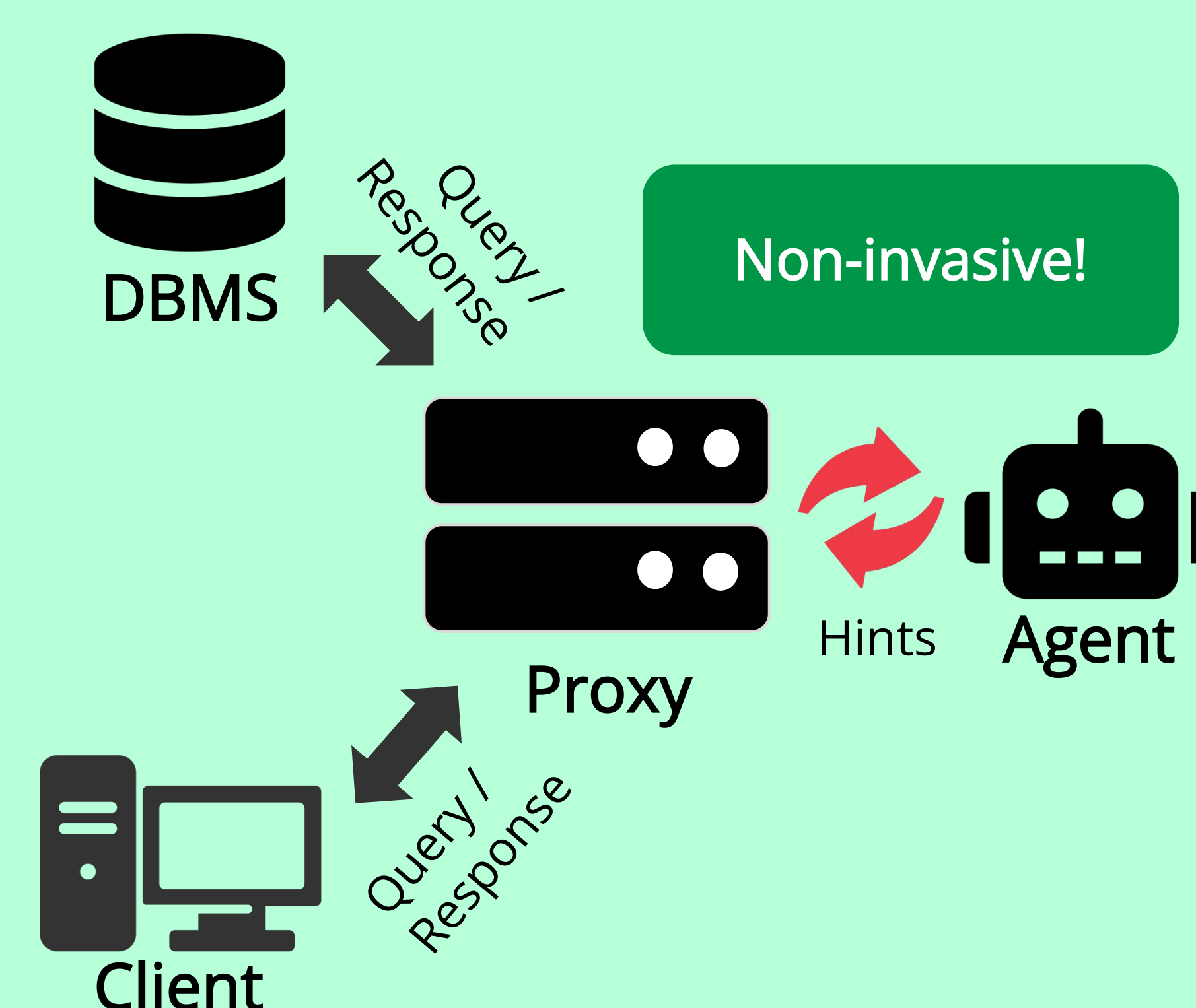
Key Idea

(1) What: LLM reasoning
(2) How: Proxy architecture

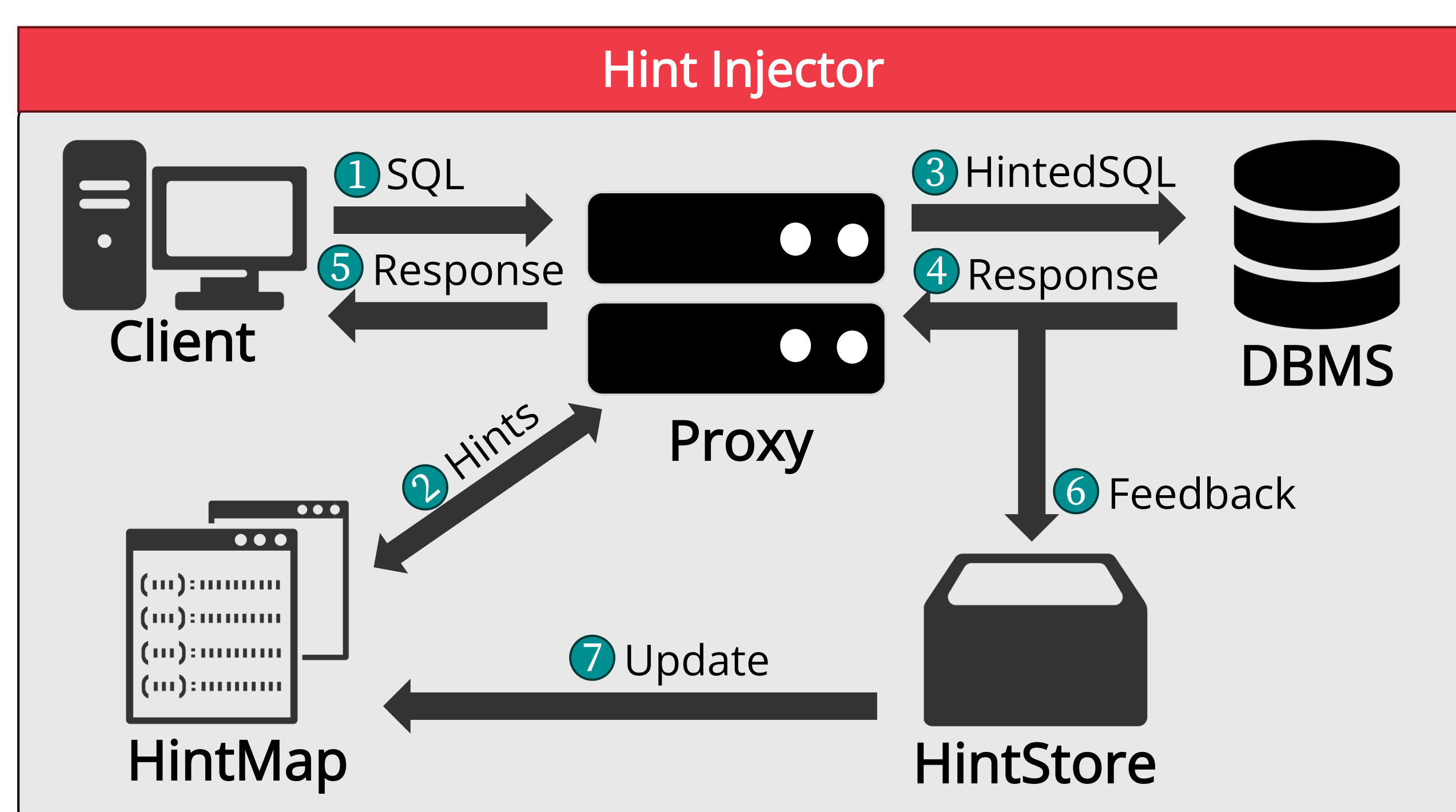


Solution

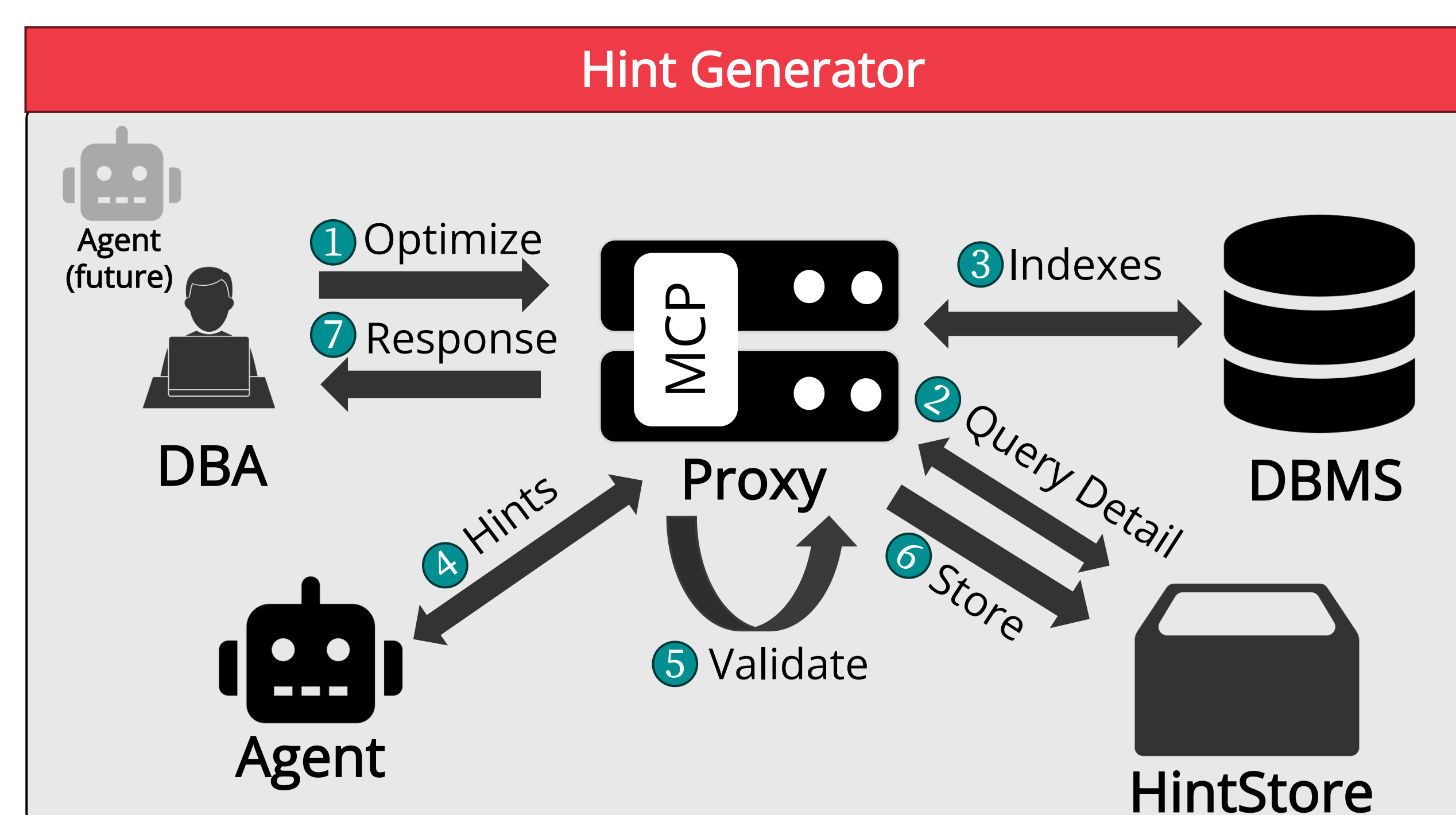
Proxy-based LLM Query Tuner



Architecture



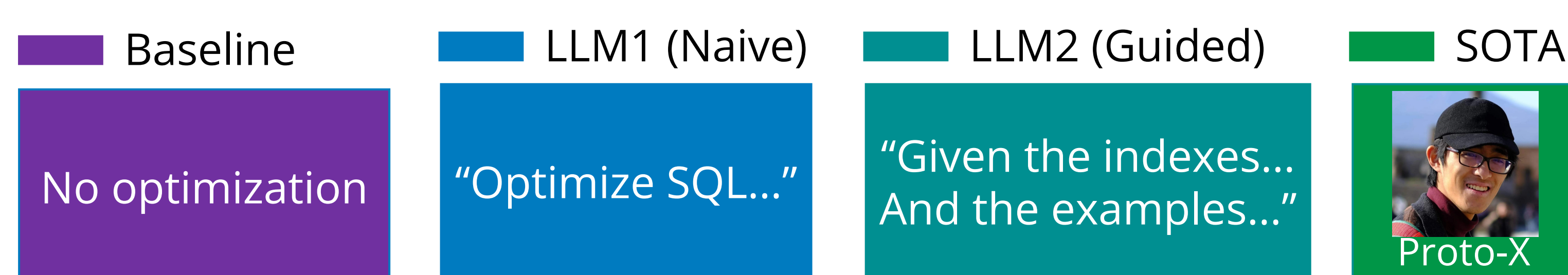
Proxy **rewrites** queries with their best matching hint



Agent **generates** new hints for slow queries

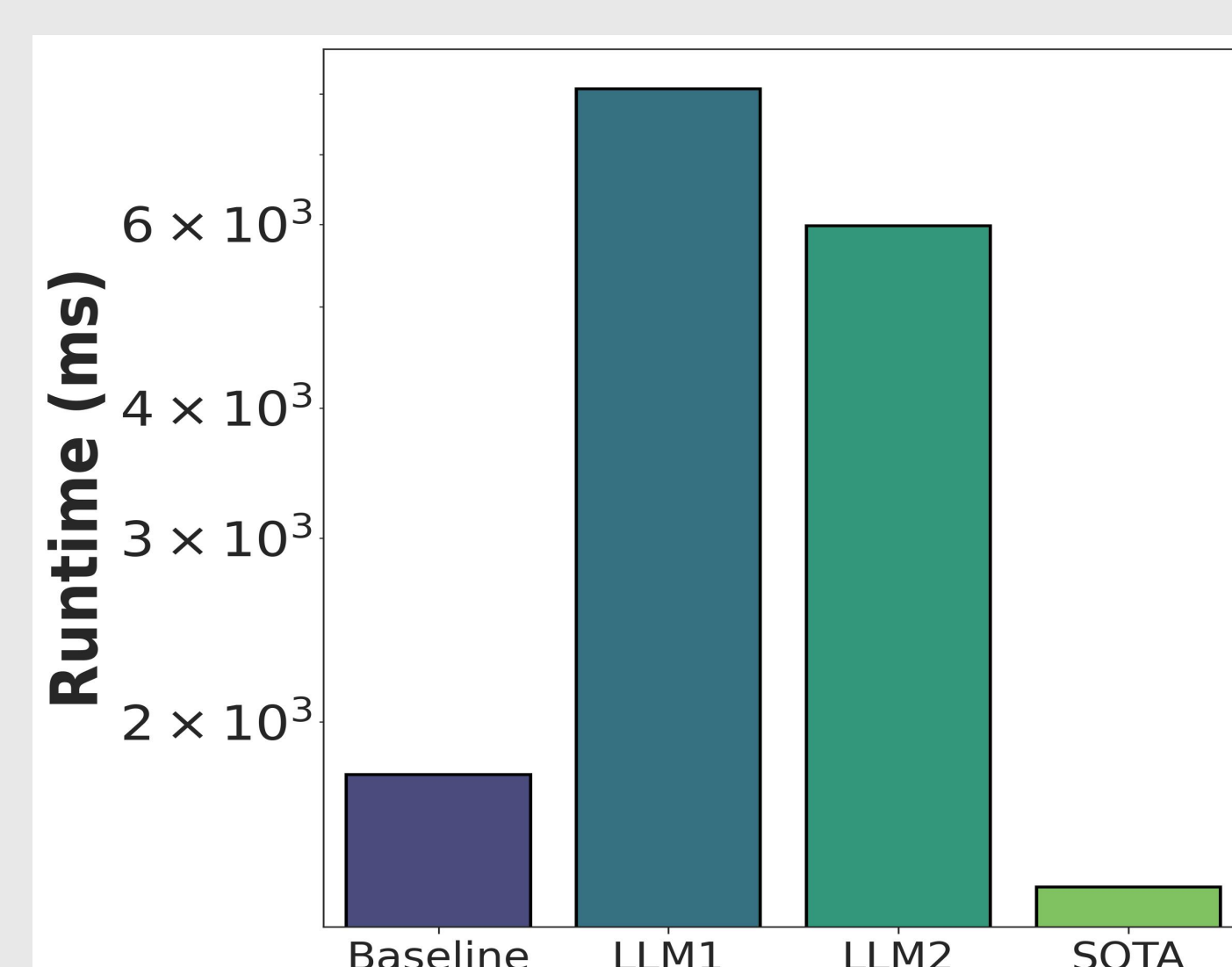
Evaluation

Benchmark: TPC-H (SF 10) on PostgreSQL 15



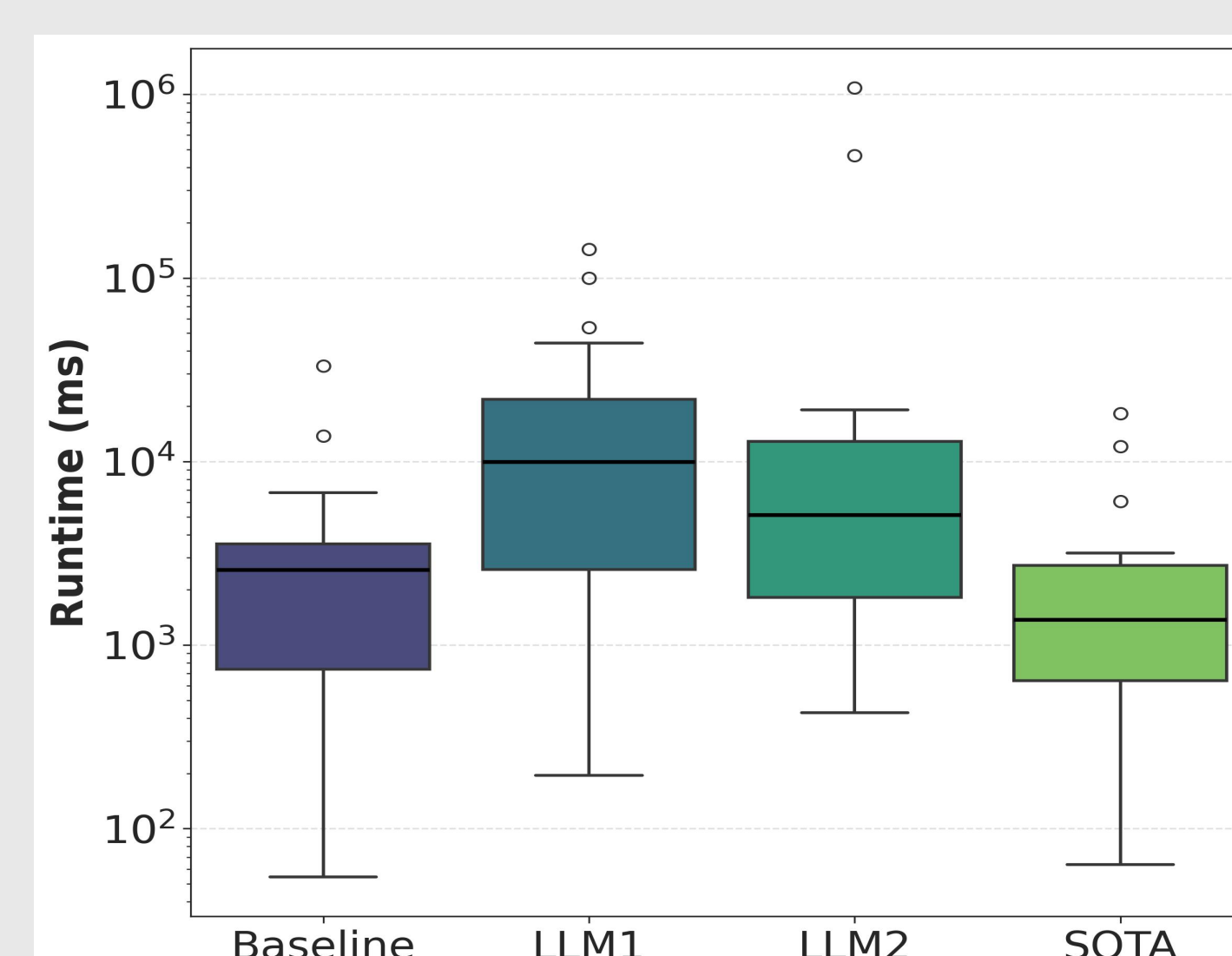
TPC-H Workload Runtime

Geometric mean of query runtimes
(lower is better)



TPC-H Per-Query Runtime

Distribution of individual query runtimes
(lower is better)



Work-In-Progress

Feedback-Driven Agent Optimization:
Continuously refine Agent behavior with feedback

Scalability and Generalization:
Extend to more workloads and more database systems

Adaptive Query Hinting Strategy:
Use reinforcement learning to prioritize queries to hint

Takeaways

Proxy-based Tuning: non-invasive design

LLM-based Reasoning: currently worse than RL-based SOTA, working on it

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